Technical Appendix

I searched and downloaded open datasets from UK government/official statistics portals (IMD, Census/ONS tables, fuel poverty), city/public-sector open data (MPS crime), and published indices/datasets (DPI, DERI, broadband availability/performance proxies), then cleaned and merged them into one LSOA-level analysis table.

1.1 Data

All analysis is at LSOA level. After joining boundaries and indicator tables, the base geography contains ~4,994 LSOAs; the complete-case modelling sample used for multivariate steps is n=4,813 (≈96%), with the remainder dropped because at least one modelling input is missing. Indicators are grouped into: (i) digital participation (DPI), (ii) socio-economic pressure/vulnerability (IMD and related indicators such as education/skills, disability/health proxies, affordability pressure such as fuel poverty, and safety context such as crime), and (iii) basic broadband performance (download/upload proxies). Variables are direction-aligned so higher values represent higher risk/constraint where needed (or explicitly inverted).

Datasets are harmonised to a single key (lsoa_code, using an LSOA11→LSOA21 lookup when required) and merged into a wide analysis table. Variables with substantial missingness are excluded from PCA, and the multivariate run uses a complete-case sample. For modelling, inputs are standardised to z-scores; heavy-tailed indicators (e.g., crime intensity) are stabilised using log1p before standardisation to reduce outlier leverage. For clustering, the run primarily reuses the PCA-standardised matrix; if missing values remain in a fallback path, they are median-imputed to stabilise k-means.

1.2 Dimensionality reduction: PCA

To avoid telling 30 separate indicator stories, I applied PCA to a standardised indicator matrix (n=4,813; 30 features). The scree plot shows a steep drop after the first components; the exported summary indicates PC1≈0.38, PC2≈0.14, PC3≈0.12 (proportion of variance), supporting a low-dimensional “story axis” interpretation for

communication. Loadings show PC1 is dominated by deprivation and education disadvantage (e.g., no qualifications and IMD components), consistent with an "everyday pressure" axis. PC2 separates a different mix of health/affordability signals and broadband-related terms, capturing additional constraints that can vary even among places with similar deprivation. PCA scores locate each LSOA in reduced space and provide the representation used for clustering.

PC1 top (+)	PC1 top (-)	PC2 top (+)	PC2 top (-)
pct_no_qual (+0.263)	crime_rate_per_1000_window (+0.014)	pct_no_long_term_condition (+0.412)	crime_rate_per_1000_window (-0.049)
risk_edu_low (+0.259)	crime_count_window (+0.012)	risk_fuel_poverty (+0.271)	rDERI_score_0_10 (-0.066)
pct_low_skill (+0.259)	bb_q_sfu_core (+0.001)	fuel_poverty_pct (+0.271)	imd_employ_score (-0.098)
imd_employ_score (+0.254)	bb_q_ul_core (-0.009)	pct_low_skill (+0.169)	imd_health_score (-0.115)
imd_income_score (+0.254)	imd_livenv_score (-0.025)	risk_edu_low (+0.169)	pct_level4plus (-0.144)
imd_score (+0.252)	bb_q_dl_core (-0.081)	pct_no_qual (+0.166)	pct_limited_a_lot (-0.209)
rDERI_score_0_10 (+0.250)	pct_no_long_term_condition (-0.119)	imd_barriers_score (+0.144)	pct_long_term_not_limited (-0.288)
imd_edu_score (+0.246)	dpi_score (-0.119)	risk_edu_highskill_inverse (+0.144)	pct_disabled (-0.312)
pct_limited_a_lot (+0.235)	pct_long_term_not_limited (-0.167)	dpi_score (+0.115)	risk_disability_limited (-0.312)
risk_edu_highskill_inverse (+0.233)	pct_level4plus (-0.233)	bb_q_ul_core (+0.092)	pct_limited_a_little (-0.365)

Figure 1. Top loadings on principal components (PC1 and PC2).

For communication (rather than a universal standard), I operationalises a "missing middle" signal using London-relative thresholds: DPI bottom 10% = 0.938 and a "good broadband" benchmark p75 download = 146.7 Mbps (used in the DPI–speed comparison figure).

1.3 K-means clustering

Neighbourhoods are grouped into k=6 recurring types using k-means in PCA score space. Clustering uses the top-15 PCA scores (explained variance ≈0.964) to preserve multi-dimensional structure while reducing noise/collinearity. Key settings: k=6 (fixed), random_state=42, n_init=20. A sampled silhouette score is reported as a rough separation check (not for selecting k), because the typology is intended for interpretation and policy communication rather than optimising clustering metric. One very small cluster C6 is treated as an outlier-type rather than a generalisable London-wide class.

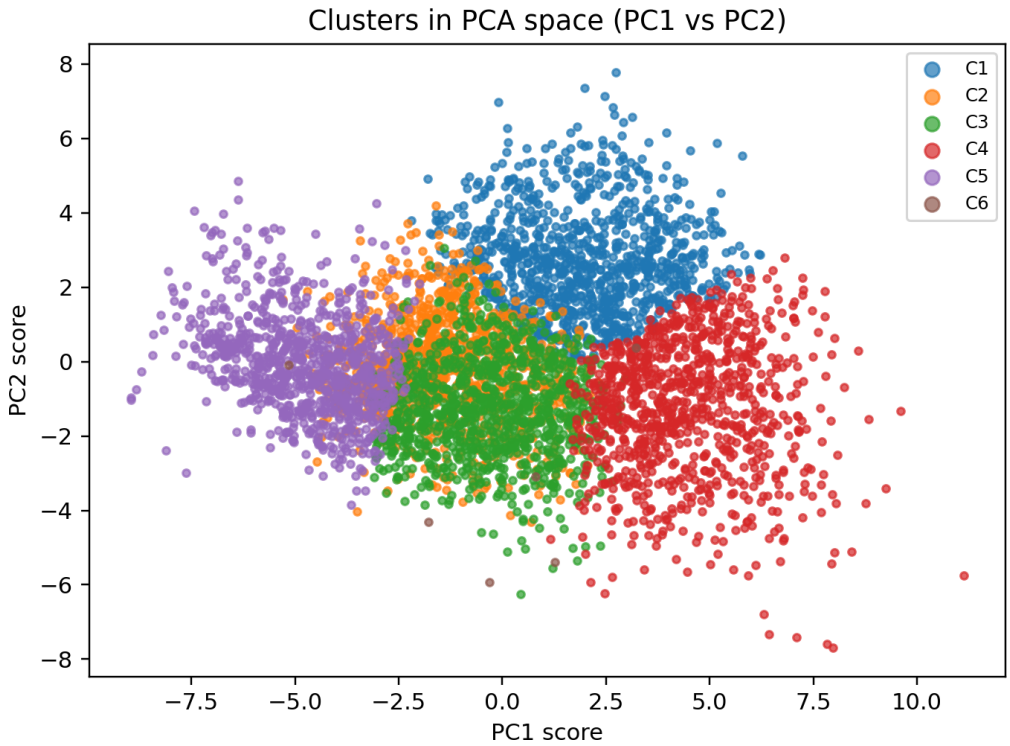

Figure 2. Clusters in PCA space (PC1 vs PC2).

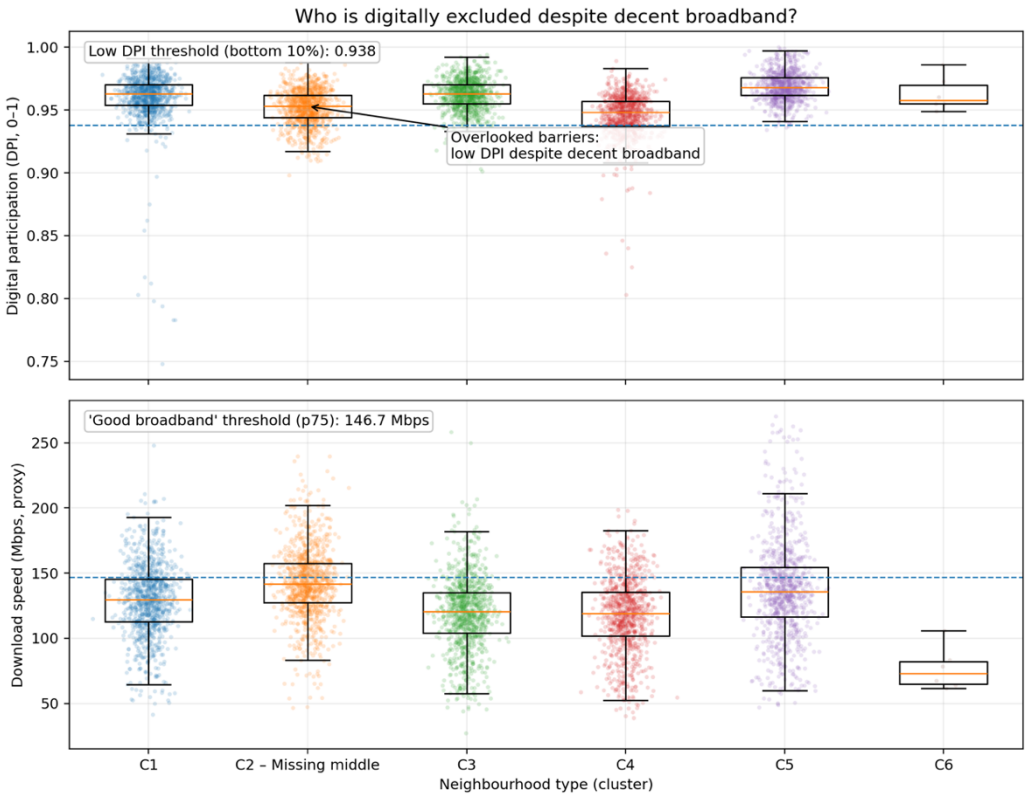

Figure 3. Who is digitally excluded despite decent broadband?

1.4 Actionable ranking: ICS risk score

To translate typology ("how places differ") into action ("where to start"), I built an ICS digital exclusion risk score combining three ingredients: (1) participation risk (DPI), (2) pressure/vulnerability, and (3) digital constraints (broadband proxies).

$$ICS = w_P \cdot participation_risk + w_V \cdot vulnerability_risk + w_I \cdot infra_risk$$

Each component is computed as the mean of risk-oriented z-scores (direction-flipped where needed) and combined with weights wP=0.45, wV=0.45, wI=0.10. LSOAs are ranked and binned into policy bands (top 10%, 10–25%, 25–50%, 50–100%); the top decile is flagged as priority. This is not simply a "worst broadband" map: the top-risk band contains both stacked-disadvantage areas and a meaningful minority of missing-middle areas—stronger broadband yet low participation—likely to be missed by connectivity-only targeting.

1.5 Benchmarking against DERI+ "overlooked" areas

ICS is compared with DERI as a benchmark sanity check. The ICS–DERI scatter is split into four quadrants using high-risk cut-offs (top-decile style thresholds): High–High (consensus high risk), Low–Low (consensus low risk), and two disagreement zones. The most policy-relevant is High ICS & Low DERI ("overlooked"): neighbourhoods flagged by the composite but not by the benchmark. Diagnostics (percentile correlation and top-k overlap/Jaccard-style agreement) show broad agreement overall but weaker alignment at the sharp end, consistent with the main-text claim that some high-risk neighbourhoods can slip through single-lens frameworks. Overlooked areas are exported for follow-up targeting and qualitative checks.

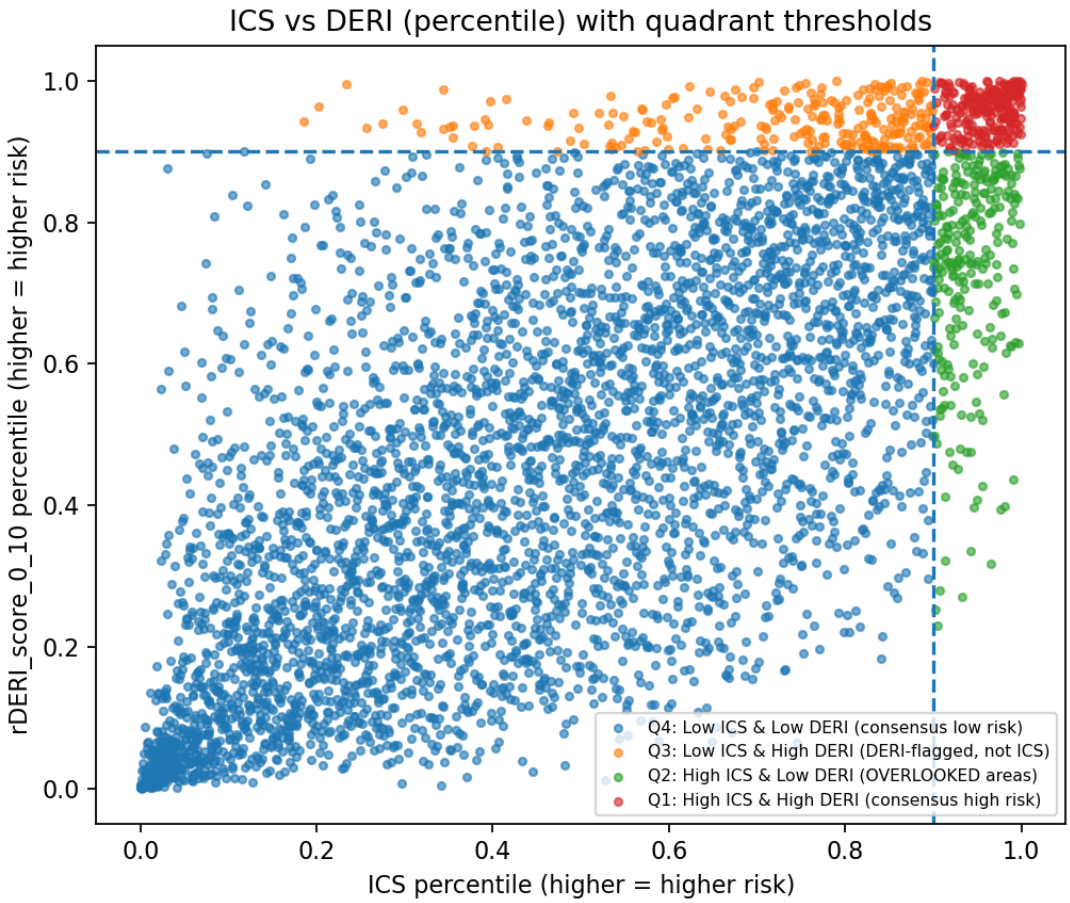

Figure 4. ICS vs DERI (percentile) with quadrant thresholds

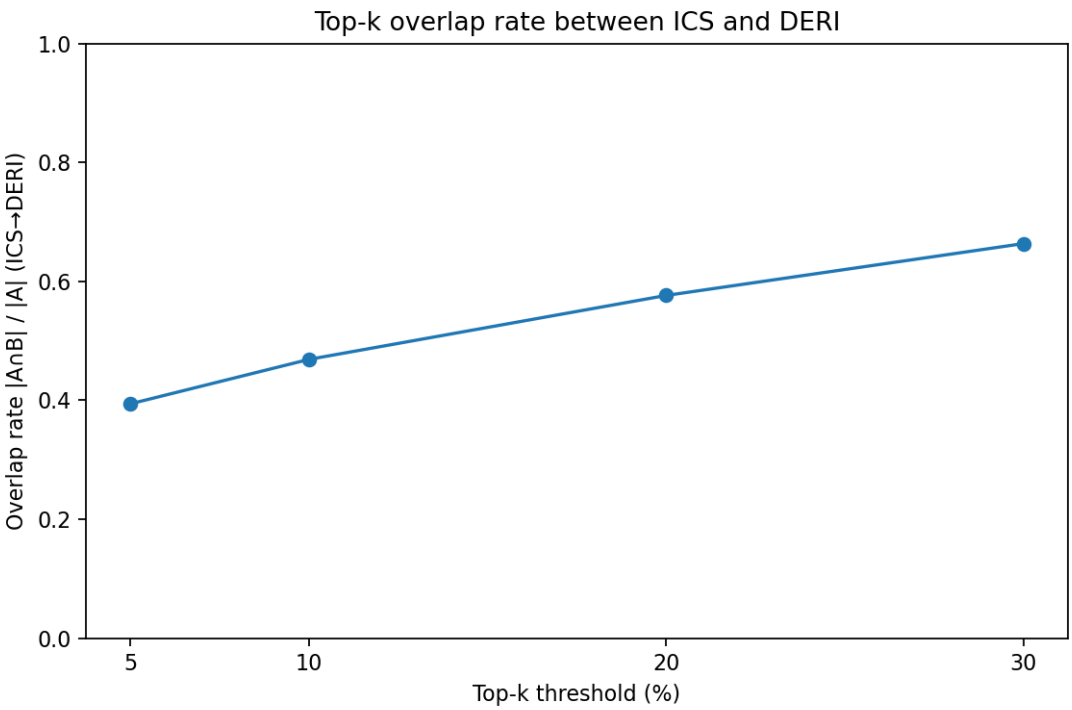

Figure 5. Top-k overlap rate between ICS and DERI

1.6 Spatial statistics: clustering is not random

To test whether risk forms neighbourhood patches rather than isolated dots, I ran spatial statistics on DPI and ICS. Global Moran's I uses Queen contiguity (polygon touches) and permutation inference (n=99); the exported summary reports DPI I≈0.405 and ICS I≈0.364 with p_sim≈0.01, supporting non-random spatial clustering. Local tests then identify where clustering occurs: Gi* maps statistically significant hotspots/coldspots of ICS, while LISA classifies local patterns (HH/LL/HL/LH). These outputs underpin the article's claim that exclusion concentrates into contiguous patches, motivating place-based rather than one-off interventions.

Figure 6. Global Moran's I for DPI and ICS

Metric	Moran's I	p-value	Permutations
DPI Moran's I	0.405	0.01	99
ICS Moran's I	0.364	0.01	99

Figure 7. Cluster profiles and overlooked rates (Q2 areas)

cluster	n_lsoa	share_%	n_overlooked (Q2)	overlooked rate_%	risk_participation (Z)	risk_vulnerability (Z)	risk_infra (Z)	risk_overall_mean (Z)
C4	910	18.9	155	17.0	1.70	0.69	0.17	0.85
C2	1026	21.3	46	4.5	0.87	-0.39	-0.31	0.05
C1	983	20.4	29	3.0	-0.25	0.37	-0.13	-0.00
C3	988	20.5	26	2.6	-0.40	0.05	0.32	-0.01
C6	6	0.1	0	0.0	-0.57	4.40	2.29	2.04
C5	900	18.7	0	0.0	-1.34	-0.74	-0.04	-0.71

1.7 Reflection and limitations.

Quantifying what it means to be "left behind by digital services" is constrained by data availability: major platforms rarely publish service-use data that can be disaggregated to London small areas (LSOAs). This study therefore relies on composite indices (DPI, DERI) as proxies for participation and exclusion. However, these indices may already include socio-economic inputs (e.g., poverty proxies) that overlap with covariates used here (notably IMD-related indicators), creating potential double-counting and robustness risks. Differences in time coverage, scaling, and indicator definitions across datasets can also shift rankings and typologies. The map should be treated as a

prioritisation tool to target resources and guide local follow-up, not a replacement for contextual judgement.

Indicators are also necessarily proxies: broadband performance does not capture affordability, device access, trust, skills, or service design quality—hence the substantive importance of the missing middle. Future work would be strengthened by finer-grained behavioural measures closer to actual service use (e.g., food-delivery density, ride-hailing trips, or small-area logs for online health/government services).